

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	AI&ML/2024	Date:	08-08-2024

Code of Conduct

I, K.Mounith(192425090) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K.Mounith

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1

Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. A company is developing an AI-powered customer support chatbot for banking services. The chatbot uses Context-Free Grammar (CFG) for parsing user queries. However, it faces issues such as:

Incorrect interpretation of ambiguous sentences

Failure to handle subject–verb agreement

Inefficient parsing for long queries Example

user query:

“Show me the transactions with the card from last month”

- Analyze the ambiguity present in the given sentence using CFG
- Evaluate the limitations of CFG in handling real-world conversational queries
- Evaluate and propose an improved solution using PCFG, feature structures, or Earley parsing
- Justify how your proposed method improves accuracy and efficiency in an industry setting

2. A voice assistant system (like Alexa/Siri-type applications) processes spoken commands using parsing techniques. The system currently uses top-down parsing, but faces:

Backtracking issues

Poor handling of incomplete or ambiguous input

Difficulty in real-time response Example

command:

“Book a flight to Delhi with a window seat”

- Analyze how the given sentence can lead to multiple parse structures
- Analyze the limitations of top-down parsing in this real-time application
- Analyze how Earley parsing can improve handling of ambiguity and partial inputs
- Compare and analyze the performance of top-down vs Earley parsing for such industry use cases

3. A healthcare company is developing an AI system to analyze patient medical reports and automatically extract critical information such as diagnosis, prescribed treatments, and follow-up actions.

The system must handle:

Complex and lengthy medical sentences

Ambiguity in sentence structure

Accurate subject-verb agreement

Relationships between medical actions and entities (sub-categorization) Real-

time processing for hospital use Example input:

"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."

The current system, based only on basic Context-Free Grammar (CFG), fails to correctly interpret relationships and produces inconsistent outputs.

a) Design a complete NLP architecture integrating:

CFG for syntactic structure

Probabilistic CFG (PCFG) for ambiguity resolution

Feature Structures for agreement handling

Efficient parsing techniques

b) Develop a step-by-step workflow showing how the system processes the given medical sentence from input to structured output (diagnosis, action, location). c) Create a method to:

Incorporate sub-categorization frames for medical verbs

Handle real-time data processing

Improve accuracy and scalability in a hospital environment

Q1. Banking Customer Support Chatbot

The chatbot uses CFG but has problems with ambiguity, subject–verb agreement, and long queries. The example is:

“Show me the transactions with the card from last month”

a) Ambiguity using CFG

- The ambiguity mainly comes from the phrase: **“with the card”**
- It can be interpreted in different ways.

Interpretation 1:

The user wants transactions that were made **using the card**.

[transactions [with the card]] **Interpretation**

2:

The phrase may be interpreted as referring to transactions associated with **a card**, creating uncertainty about which entity “with the card” modifies.

There is also potential ambiguity regarding:

“from last month”

It may specify the time period of the transactions, which is the most natural interpretation, but a weak grammar may attach it incorrectly to another phrase.

CFG can generate multiple valid parse trees without necessarily knowing which interpretation the user intended.

b) Limitations of CFG

CFG is useful for representing basic syntactic structures, but real conversational language is more complex.

Major limitations include:

- It does not naturally represent probabilities.
- It has difficulty resolving semantic ambiguity.

- Basic CFG does not enforce subject–verb agreement.
- Long sentences can produce many possible parse trees.
- It does not use conversational context effectively.
- Different meanings may have the same syntactic structure.

Therefore, a banking chatbot based only on CFG may produce grammatically possible but semantically incorrect interpretations.

c) Improved solution

A better solution is to combine PCFG, feature structures, and efficient parsing.

PCFG assigns probabilities to alternative parse trees and can select the most likely interpretation.

Feature structures can represent grammatical information such as:

Number = Singular/Plural

Person = 1st/2nd/3rd

This helps enforce agreement.

Earley parsing is useful because it can handle a broad class of CFGs and is suitable for ambiguous or incomplete inputs.

A practical architecture can be:

User Query → Tokenization → Earley Parser → PCFG Ranking → Feature Checking → Semantic Interpretation → Banking Action

d) Industry benefits

The improved system can:

- Resolve ambiguous interpretations more effectively.
- Handle long queries more efficiently.
- Enforce grammatical constraints.
- Rank competing interpretations.
- Produce more accurate banking actions.

For example, it can convert the query into a structured representation such as:

QUERY(Transactions, Card, Time=LastMonth)

This representation can then be passed to the banking service.

Conclusion: A combination of Earley parsing + PCFG + feature structures is more suitable than basic CFG for a real banking chatbot because it combines efficient parsing, ambiguity resolution, and grammatical control.

Q2. Voice Assistant – Flight Booking

The voice assistant receives:

“Book a flight to Delhi with a window seat”

The existing system uses top-down parsing and has backtracking and real-time-response problems. a) Multiple parse structures

The main ambiguity concerns:

“with a window seat”

The phrase can be associated with the flight booking request as a whole, meaning:

Book [a flight to Delhi] [with a window seat]

The intended interpretation is that the passenger wants a flight to Delhi and wants a window seat. A parser may also temporarily consider alternative attachment structures while processing the sentence, especially if the grammar contains several possible noun-phrase or prepositional-phrase attachments.

The semantic interpretation should ultimately be: BookFlight(destination=Delhi, seat=Window)

b) Limitations of top-down parsing

Top-down parsing starts from the start symbol and predicts possible structures.

In a voice assistant, this can cause:

- Extensive backtracking.
- Repeated exploration of incorrect grammar paths.
- Slow processing when ambiguity increases.
- Difficulty with incomplete speech.
- Poor real-time performance.

For example, while processing spoken input, the system may not yet know the complete sentence but may already need to begin interpretation.

c) Earley parsing

Earley parsing is better suited for ambiguous and incomplete input because it maintains multiple possible states during parsing rather than repeatedly restarting failed parsing paths.

It can:

- Handle ambiguous grammar.
- Process partial input.
- Avoid unnecessary repeated backtracking.
- Maintain multiple possible interpretations.
- Work with general CFGs.

For the example, Earley parsing can maintain the interpretation of the flight destination while continuing to process the seat preference.

d) Comparison

Feature	Top-Down Parsing	Earley Parsing
Ambiguity handling	Less efficient	Better
Backtracking	Can be high	Reduced
Partial input	Difficult	Better
General CFG support	Depends on parser	Yes
Real-time suitability	Limited	More suitable
Multiple parse states	Less flexible	Explicitly maintained

Conclusion: For a real-time voice assistant, Earley parsing is more appropriate because spoken input can be incomplete, ambiguous, and continuously generated.

Q3. Healthcare NLP Semantic Processing Architecture

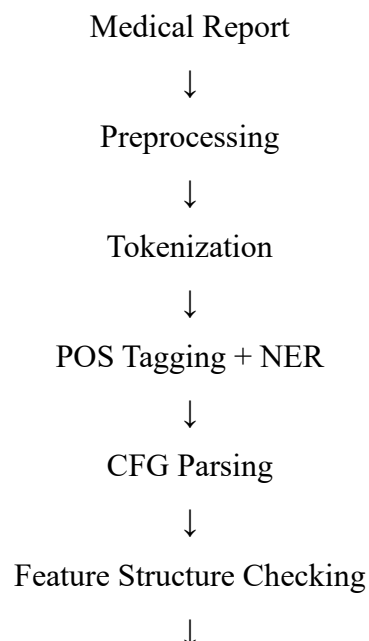
The medical sentence is:

“The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai.”

The existing CFG-based system produces inconsistent relationships.

a) Complete NLP architecture A

suitable architecture is:



PCFG Ambiguity Resolution



Subcategorization Analysis



Semantic Role Labeling



Structured Medical Information CFG

provides the basic syntactic structure.

PCFG assigns probabilities to possible interpretations and helps select the most likely parse.

Feature structures represent grammatical features such as number, person, and agreement.

Efficient parsing improves processing speed, which is important in hospitals.

b) Step-by-step processing Step 1 – Input

The system receives the medical sentence.

Step 2 – Tokenization

The sentence is divided into tokens such as:

doctor, reviewed, patient, recommends, medication, follow-up, Chennai.

Step 3 – Entity recognition The

system identifies:

- Doctor → Medical professional
- Patient → Patient entity
- Medication → Treatment
- Chennai → Location

Step 4 – CFG parsing

The system identifies the main syntactic structure:

- Doctor → recommends → starting medication and:
- Doctor → recommends → scheduling follow-up visit

The relative clause:

who reviewed the patient last week is associated with the doctor.

Step 5 – Feature checking

The system verifies agreement: •

doctor → recommends

- The singular subject agrees with the singular verb.

Step 6 – PCFG

If several possible parses exist, PCFG assigns probabilities and selects the most likely structure.

Step 7 – Semantic extraction

The final structured information can be represented as:

- Agent: Doctor
- Previous Action: Reviewed Patient
- Treatment Action: Start Medication
- Follow-up Action: Schedule Visit
- Location: Chennai
- Time: Last Week

c) Subcategorization frames

Medical verbs can be associated with expected arguments.

For example:

- recommend + action
- review + patient
- prescribe + medication + patient For the given sentence:
- recommend (Doctor, starting medication)
- recommend (Doctor, scheduling follow-up visit)

The system should maintain medical verb frames so that it knows which entities and actions can naturally occur with each verb.

Real-time processing and scalability

For hospital use, the system should use:

- Efficient parsing algorithms.
- Caching of common medical terms.
- Parallel processing where appropriate.
- Domain-specific language models.
- Structured medical dictionaries.
- Confidence scores.
- Error detection and validation.

Conclusion: Combining CFG, PCFG, feature structures, subcategorization frames, and efficient parsing provide a much stronger healthcare NLP system than relying on basic CFG alone.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
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1								
2								
3								

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____